

SEN ZHANG

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SUMMARY

Registered NY architect and computational design technologist. Build and maintain firm-scale design-automation platforms — Revit/Rhino plugins, project-analytics dashboards, on-prem RAG infrastructure, and AIA 2030 reporting pipelines. Bridge architectural intent and shippable code.

EXPERIENCE

Ennead Architects LLP (now Ennead, a CannonDesign company)

Computational BIM Developer · 2022–Present **BIM Manager (concurrent)** · 2022–Present **Rhino Doctor (concurrent)** · 2022–Present **Architectural Designer, BIM Specialist** · 2020–2022 New York, NY

- Architect and lead developer of EnneadTab — 450+ tool platform spanning Revit, Rhino, CAD, Adobe, Excel, Email, and LLM integration; 3 U.S. Copyright registrations
- BimRunner — Revit project-health analytics dashboard (Next.js + Vercel Postgres, ~91k LOC, 422 files)
- EnneadTabLLM — local-deployment Ollama RAG monorepo (Windows server + Vercel proxy + shared TypeScript types) for on-prem AI workflows
- AIA-Sustainable-Reporting — CRON Excel pipeline + frontend dashboard for AIA 2030 metrics
- FieldDay — offline-first PWA for site inspection on phone/tablet/desktop with PDF spatial pinning
- BIM Manager: NYU Langone, Northwell, FTZ Shanghai, ByteDance Houhai HQ, Bilibili HQ, Lenox Hill
- Rhino Doctor: Rhino-to-Revit interop with zero-learning-curve internal tooling; complex parametric and industrial-scale geometry triage
- Speaker representing Ennead Architects, Design Technology at Autodesk University 2024 (San Diego) and Advancing Computational Design Conference 2026 (Austin)

Arquitectonica

Designer, BIM Specialist, CG Visualization · 2017–2020 New York, NY

- 40 Hudson Yards (NYC office tower), Amazon HQ2 master planning (Chicago), Toronto SmartCity 31 Parliament
- Curtis Block (Calgary), Warsaw Hotel (Poland), Newport Waterfront, Equinox Hotel (San Jose), Queens Sherbourne mixed-use (Toronto)
- Residential, office, hotel, and master-planning projects with end-to-end BIM coordination and CG visualization delivery

RPI Publication Dept.

Book-design, Asset Management · 2016–2017 Troy, NY

Terreform ONE

Research, Design · 2016–2017 New York, NY

SELECTED PROJECTS — DESIGN TECHNOLOGY

PROJECT	ROLE	PERIOD	DESCRIPTION
EnneadTab	Architect & Lead Developer	2021–Present	Cross-application architectural plugin ecosystem (Revit, Rhino, CAD, Autodesk Desktop Connector, Enscape, InDesign, Photoshop, Excel, Email, AI/LLM, TextToSpeech). 450+ tools, deployed firm-wide, three U.S. Copyright registrations. CPython compile pipeline → distribution repo → user-side scheduled updater with delayed-deletion silent background updates. Cross-application architectural plugin Eco-system that span Rhino, Revit, CAD, Autodesk Desktop Connector, Enscape, InDesign, Photoshop, Excel, Email, Notification, AI, Computer Resource Monitoring, TextToSpeech, Sound Effect.
BimRunner	Architect & Full-Stack Developer	2026	Revit project-health analytics dashboard. Next.js 14 + Vercel Postgres + Chart.js. ~91k LOC across 422 files. RESTful API for data ingestion, dynamic scoring from stored facts, file-attachment extensibility.
EnneadTabLLM	Architect & Developer	2025–2026	Ollama-based local Retrieval Augmented Generation monorepo. Windows server (Ollama + RAG pipeline) + Vercel Next.js proxy + shared TypeScript types. Solves the on-prem AI problem for HIPAA / data-sovereign clients.
AIA-Sustainable-Reporting	Architect & Lead Developer	2026	CRON service for Excel processing + frontend dashboard for AIA 2030 sustainable-design metrics. Single-source <code>.env.local</code> config across Next.js app, cron scripts, and CLI; Vercel CLI + Codespaces friendly.
FieldDay	Architect & Lead Developer	2026	EnneadTab-FieldDay — offline-first PWA for site inspection on phones, tablets, and desktops. PDF-based spatial pinning, photos, voice notes, and comments with zero-connectivity capture and Vercel Pro cloud sync. Preserves spatial accuracy across drawing revisions.
EnneadTabHome	Architect & Developer	2025–2026	Landing page and central hub for EnneadTab applications at enneadtab.com. Next.js 14, Tailwind, mobile-friendly UI.
EnneadTabWiki	Architect & Lead Author	Ongoing	Public documentation site for the EnneadTab plugin ecosystem.
Rosette	Architect & Developer	2026	Architectural-language translator: turns design-team / client / regulatory text into one-line plain-English (or Mandarin) summaries plus an interactive structural graph. Part of the EnneadTab product line.
RenderPolisher	Architect & Developer	2025–2026	Internal AI-driven render post-processing pipeline for architectural visualization workflows.
2025 Performance Review	Architect & Lead Developer	2025–2026	Firm-wide employee evaluation system. Excel ingest → interactive HTML report with software-proficiency tracking and visual analytics. Deployed to GitHub Pages.
LakeHouse-Design.com	Architect & Developer	Ongoing	Rhino package of professional design tools — dynamic area takeoff, layer/material merging, object transformation, block management.

SPEAKING & RECOGNITION

YEAR	VENUE	TITLE	ROLE
2026	Advancing Computational Design Conference, Austin, TX	<i>The Design of Design — Enabling Agile Introduction of New Tools to Scale Design Efficiency</i>	Speaker
2024	Autodesk University, San Diego, CA	<i>Revit As A Game Engine</i>	Session Speaker
2023	Digital Built Week Americas, Dallas, TX	<i>Promoting Computational Design to Non-Programmers</i>	Panel Speaker
2023	AEC Hackathon, New York, NY	<i>Educational Tool Connecting Junior and Senior Students</i>	Team — Honorable Mention

EDUCATION

- **Columbia University GSAPP** — M.S. Advanced Architecture Design — New York, NY · 2016–2017
- **Rensselaer Polytechnic Institute** — Bachelor of Architecture — Troy, NY · 2012–2016

- **Tongji University** — Major in Urban Planning — Shanghai · 2010–2012

CREDENTIALS & TECHNICAL STACK

- Registered Architect, New York (NCARB)
- Fluent: CPython, IronPython, TypeScript, JavaScript, React, Next.js
- Tooling: Revit API, RhinoCommon API, Grasshopper, Dynamo, MCP, LLM integration (Ollama, OpenAI, Anthropic)
- Languages: English, Mandarin Chinese (native)

SELECTED COMPETITION AWARDS

- EnneadTab — U.S. Copyright Office — 3 registrations, authorship credited to Sen Zhang (2024)
- Hyperloop Campus, Las Vegas — Honorable Mention (2020)
- Greenway to Ideal City, Chengdu — 2nd Place (2019)
- Tokyo Anti-Library — Honorable Mention (2018)
- Non Architecture: Dance — Finalist (2018)
- AIA Henry Adams Award (2016)
- Magna Cum Laude Honors — Dean's List 2012–2016 (2016)
- George T. Droste Scholarship — Structural engineering study (2015)
- Faculty Selection Award — Academic proficiency in Architecture (2015)